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In [1]: #Filtering the Eleemnts of ndarray
```

```
In [2]: import numpy as np
```

```
In [3]: lst=[10,-20,34,-45,67,-38,-12,0,13]
a=np.array(lst)
print(a,type(a))
```

```
[ 10 -20  34 -45  67 -38 -12   0  13] <class 'numpy.ndarray'>
```

```
In [6]: #Step-1: Prepare Boolean array with Condition--ndarray with Conditio
ba=a>0
print(ba,type(ba))
```

```
[ True False  True False  True False False False  True] <class 'numpy.ndarra
y'>
```

```
In [7]: #Step-2 Pass boolean Array to ndarray object---ndarrayobj[boolean array]
a[ba]
```

```
Out[7]: array([10, 34, 67, 13])
```

```
In [8]: #OR--Direct Approach
#Syntax: ndarrayobj[ndarrayobject with condition]
a[a>0]
```

```
Out[8]: array([10, 34, 67, 13])
```

```
In [9]: a[a<0]
```

```
Out[9]: array([-20, -45, -38, -12])
```

```
In [10]: lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
a.shape=(3,3)
print(a,type(a))
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
```

```
In [12]: #obtain Multiples of 3
a[a%3==0]
```

```
Out[12]: array([30, 60, 90])
```

```
In [13]: #obtain Multiples of 4  
a[a%4==0]
```

```
Out[13]: array([20, 40, 60, 80])
```

```
In [14]: #obtain Multiples of 5  
a[a%5==0]
```

```
Out[14]: array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
In [17]: a[(a>20) & (a<55)] # Use Bitwise Operator But not Logical Operators
```

```
Out[17]: array([30, 40, 50])
```

```
In [18]: # a[20<a>55] # Invalid
```

```
-----  
ValueError                                Traceback (most recent call last)  
~\AppData\Local\Temp\ipykernel_10440\1036895456.py in <module>  
----> 1 a[20<a>55]
```

ValueError: The truth value of an array with more than one element is ambiguous. Use a.any() or a.all()

```
In [ ]:
```